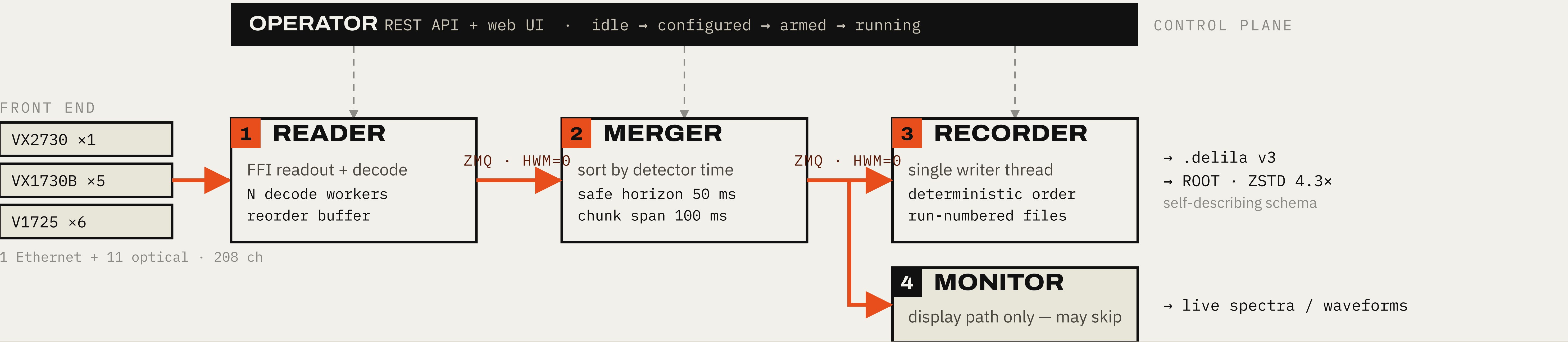


Never Drop a Hit

A triggerless, loss-free distributed DAQ in Rust. What it does under load, measured on a live fission campaign: **5.28 MHz sustained for 13 hours**, and **3.3 days** of continuous running without losing a single pulse.

S. Aogaki
ELI-NP / IFIN-HH, Măgurele, Romania · [conference, venue, date](#) [TO CONFIRM](#)

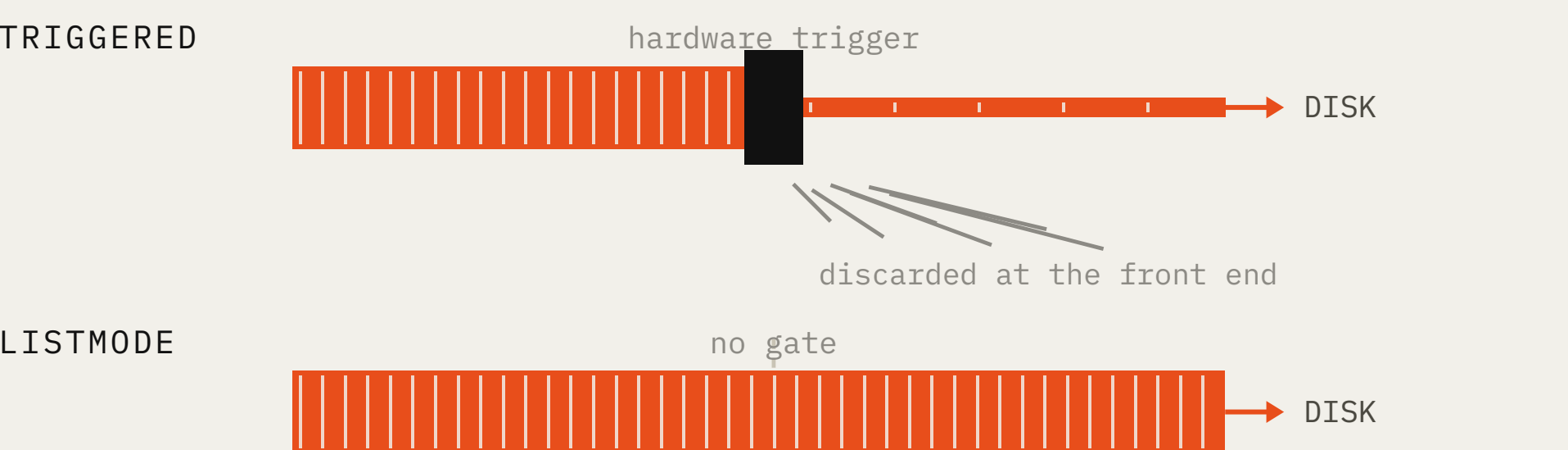
5.28 MHz	251 G	78.3 h	0
SUSTAINED EVENT RATE HELD FOR 13.2 H · RUN 0156	EVENTS IN ONE RUN 4.30 TB · 4 004 FILES · ALL VALID	LONGEST UNBROKEN RUN 3.3 DAYS · RUN 0390	PULSES MISSING OF 559 017 ON EVERY ONE OF THREE BOARDS



THE PROBLEM

No trigger, so nothing throttles the rate

Listmode: every hit written out on its own, with its own timestamp; coincidences applied later, offline.



Nothing narrows the stream. A triggered readout decides at the front end what is worth keeping. Listmode decides nothing — so the rate reaching the disk is the rate at the detector, and the only elastic element left in the chain is the buffer.

DESIGN RULE · NON-NEGOTIABLE

If the system cannot keep up, it **stops. It does not throw data away.**

- Every ZeroMQ socket runs `HWM = 0`; internal channels are unbounded. No back-pressure path is permitted to drop.
- One documented exception: hits still in flight when the run is stopped. They are counted in `dropped_batches` and logged — **never silent**. A non-zero count *while running* is a bug report, not a tuning knob.

CONCURRENCY MODEL

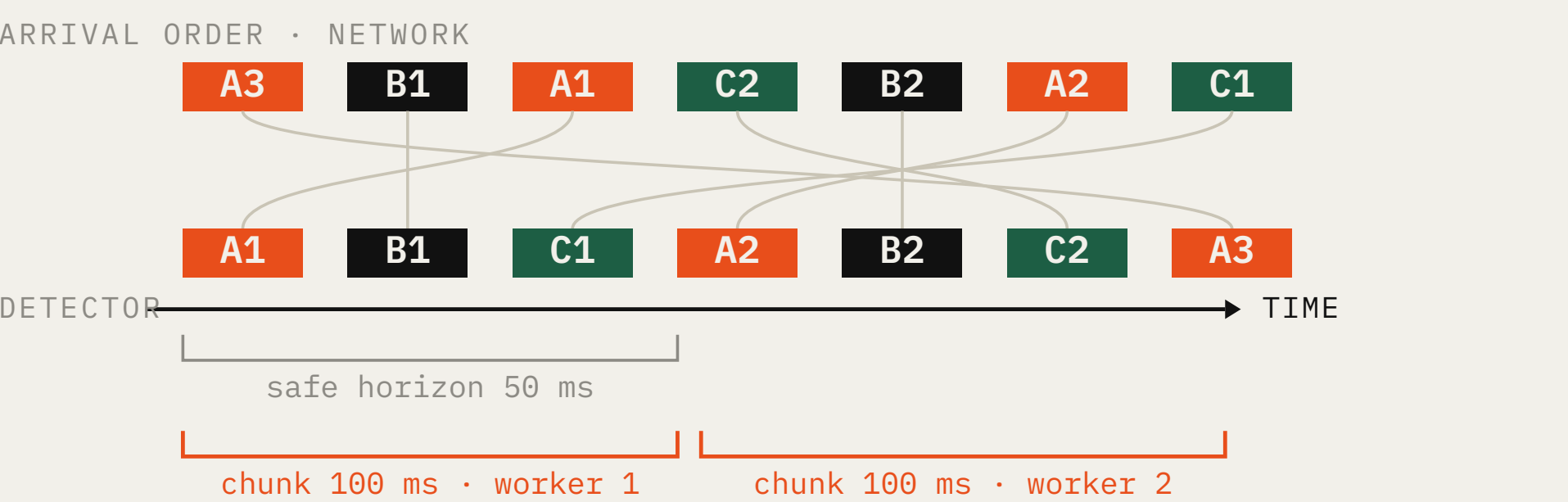
One task per stage, no shared locks

No mutex is ever held across a stage boundary, so a slow consumer cannot stall an unrelated producer. Decoding — the measured bottleneck — is fanned out over N workers behind a reorder buffer, so **the number of workers never changes the output**.

- 66 k lines of Rust, 761 automated tests
- unsafe confined to the CAEN FFI wrapper layer
- No pattern-matching heuristics in the decoder hot path — every field traced to a firmware specification page and verified against hardware

EVENT BUILDING

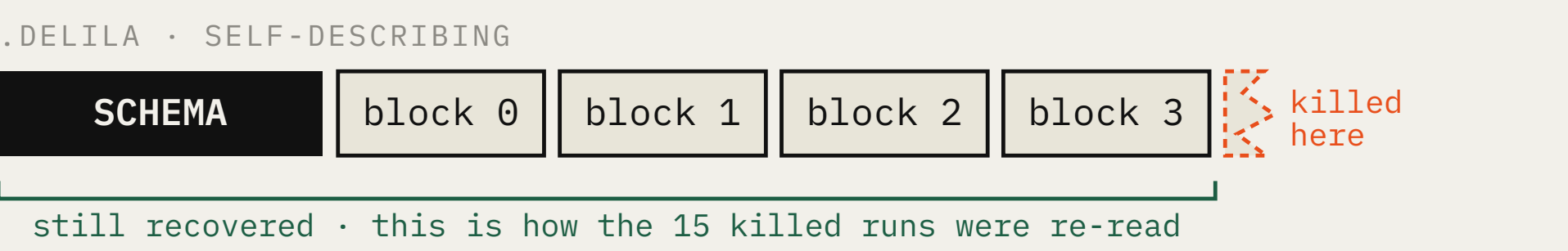
Sorted by the detector, not by the network



Three boards, one time base. Hits arrive interleaved — different links, different queue depths. A safe-horizon window bounds how far back the sorter must look; fixed-span chunks then go to pure-function workers and are re-sequenced by a single writer.

DATA AND CONTROL

Files that outlive the software



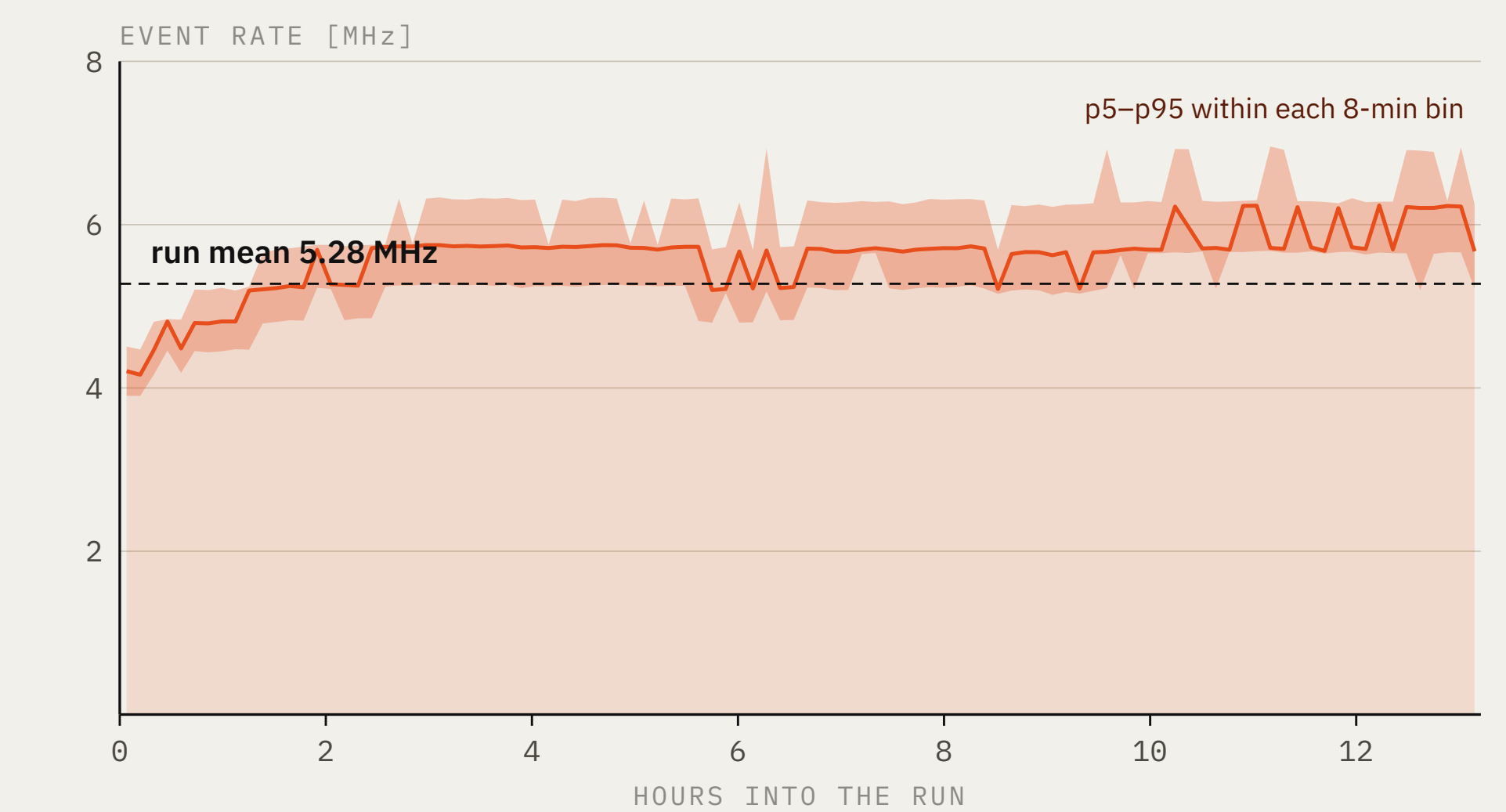
The file carries its own schema, so it stays readable without the version that wrote it. ROOT output uses native ZSTD (4.3×). A single Operator process drives the whole chain over REST and a web UI, run history in MongoDB.

UNDER LOAD · RUN 0156

Thirteen hours at 5.3 million events a second

A 60Co + 252Cf timing-alignment run across the full readout — **twelve digitizers, 208 channels**, one on Ethernet and eleven on optical link. The six DPP-PSD boards carried the rate; the V1725 Si channels sat quiet, no source in front of them. Not a benchmark — just the heaviest thing the chain has been asked to do.

Events in the run	251 023 817 095
Written to disk	4.30 TB
Duration, unbroken	13.22 h
Mean event rate	5.28 MHz
Median · peak per file	5.69 · 7.81 MHz
Time to fill one GiB	11 s
Files, and how many closed clean	4 004 / 4 004



Every one of 4 004 files, back to back. Median event rate per 7.9-minute bin, with the p5–p95 spread; each file is one GiB closed and the next opened without a gap. The rate *climbs* through the run rather than flattening, so the chain was tracking the source, not clipping against a ceiling — the write path still had headroom at the 7.81 MHz peak. At 17.1 B/event this is **90.5 MB/s** sustained for half a day.

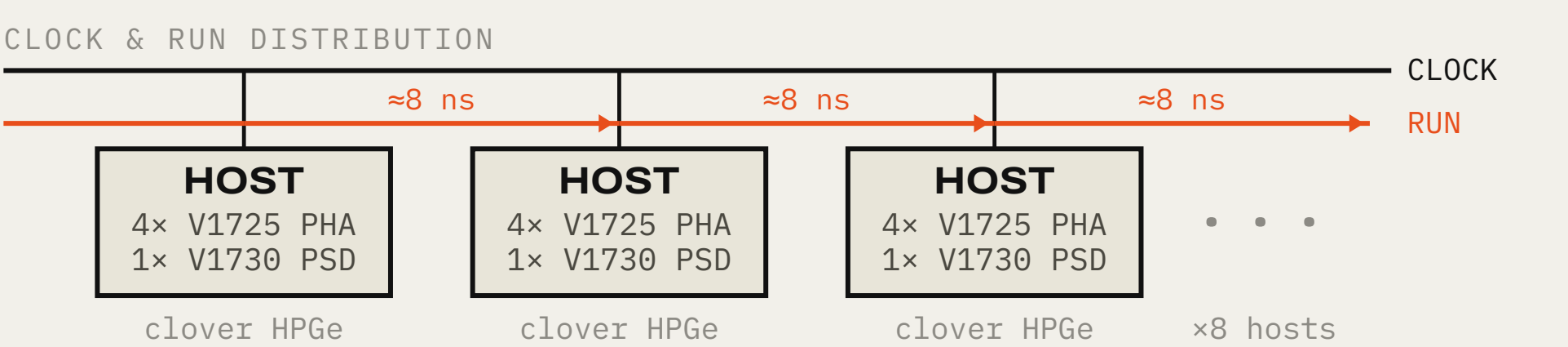
VERIFICATION

Audited, not asserted

- ✓ All 407 runs re-read from disk. 392 closed cleanly; the 15 killed mid-file were recovered by the block scanner — which is why the count above can be stated at all.
- ✓ Event-builder output checked against an independent offline analysis of the same run — coincidence yields identical, **event for event**.
- ✓ Order-inclusive hash identical for 1 and 4 worker threads — parallelism is not visible in the output.
- ✓ End-of-stream markers carry their run number, so a late marker cannot truncate the following run.

SCALING UP

ELIADE: 40 boards on one clock



y-y coincidence across clovers, built online. Eight nodes are provisioned and verified end-to-end. Ge resolution tuning runs through 2026, **beam from January 2027**.

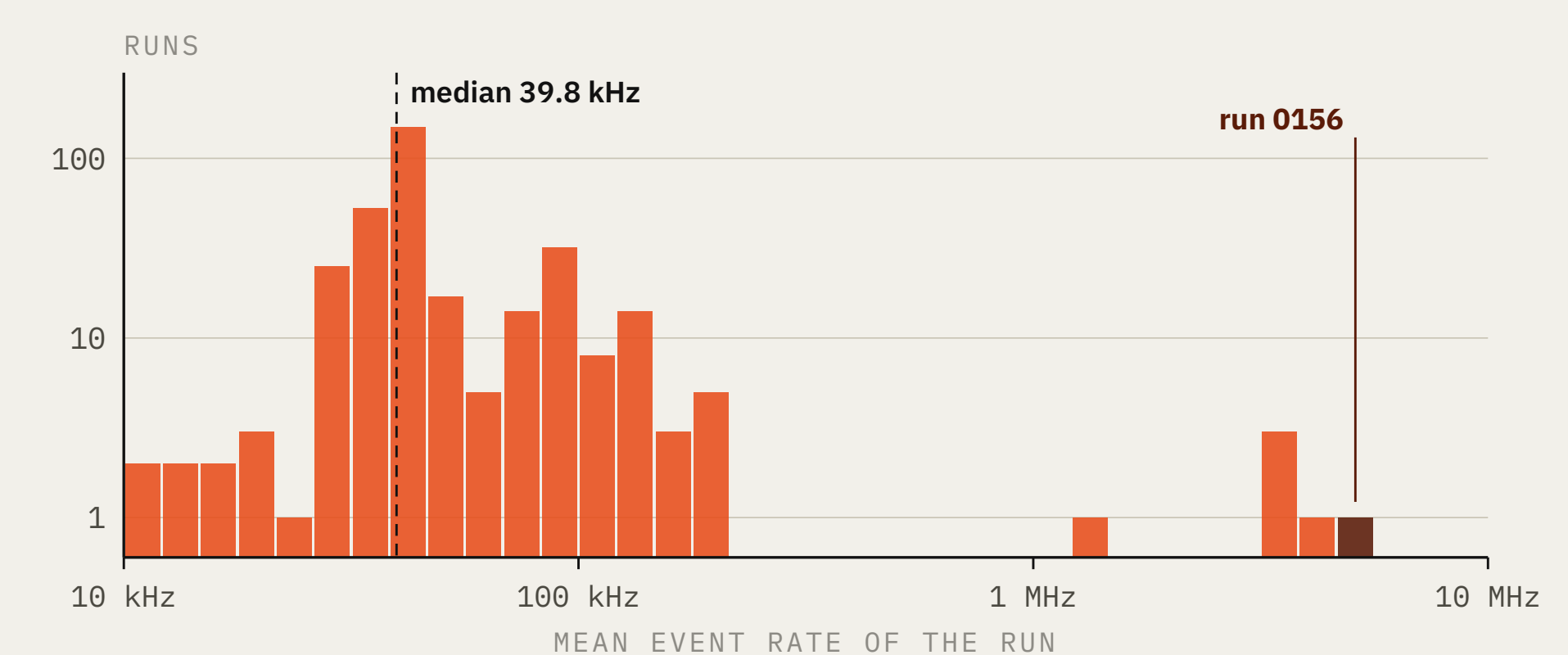
LOSS-FREE · RUN 0390

A 2 Hz heartbeat, 3.3 days long

The same campaign fans a pulser into every channel. That turns the claim into something countable: over **78.27 hours** of continuous running — twelve digitizers, 208 channels, **10 991 102 925** events — the pulser should fire **559 017** times. Every one of them has to be in the file.

BOARD	LINK	RECORDED	MISSING
VX2730	Ethernet	559 017	0
VX1730B	optical	559 017	0
VX1730B	optical	559 017	0

Three boards, two different transports, not one gap longer than a single period. The pulser runs at **1.984 Hz**, not the nominal 2 Hz — assume the round number and it looks like 0.8 % is missing. Two of the three channels also carry detector signal and show 564 *extra* hits; the third is pulser-only and shows exactly the comb.



Where the stress test sits. Mean event rate of each of the **341 runs that closed cleanly** — the whole campaign, not a chosen few. Ordinary running sits at 39.8 kHz; above 200 kHz the distribution empties for nearly a decade before a handful of runs stand an order of magnitude clear. Rates estimated from each run's first file; run0156's exact figure, over all 4 004, is 5.28 MHz.

HOW IT WAS BUILT

Specified in prose, written by agents

Nearly all of this code was produced by **AI agents working from written task documents** — one per unit of work, each carrying the requirement, the decisions and their reasons, and the evidence of hardware verification. The prose is not documentation written afterwards. It is the input.

TODO/ · 82 task docs	96,867
docs/ · 46 design docs	51,171
source comments	65,418
Rust code tokens	158,132

Counted in words, not lines or bytes. 53 % of the prose is Japanese, at **2.00 characters per word** against **6.10** for English — measure this corpus by character and its Japanese half shrinks threefold. Japanese segmented morphologically; Rust counted as identifier and literal tokens.

AVAILABILITY

Run the whole chain with **no digitizer attached**.

Reader, merger, recorder, monitor and operator all come up against a built-in emulator, so the whole pipeline runs on any machine. Every ELIADE node is accepted this way — 10 300 events, zero trigger loss — before it is trusted with a detector.

```
scripts/start_daq.sh \
config/config_emulator_local_test.toml
```

In production today. Next: a ROOT-native C++ event builder, and trapezoid-filter auto-tuning for Ge resolution.
BSD-3-Clause · github.com/ELI-NP/delila-rs